EDUBRIDGE AI · PRODUCT PITCH · 2026

Offline-first learning

for every Nigerian student

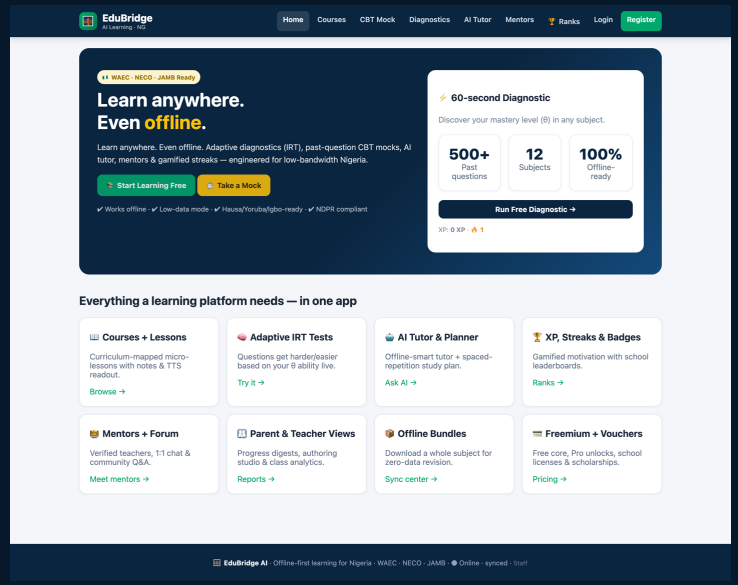
WAEC · NECO · JAMB aligned. Works on 2G, syncs when the network returns, and adapts to each learner's ability — with a free core that never locks a student out.

499
QUESTION BANK

26
SUBJECTS

31
TOPICS

5
LEVELS



Live product · EduBridge AI

THE PROBLEM

Nigeria's learners are online only on paper

Bandwidth cost, power intermittency and exam-prep quality make 'just put it online' fail.



Data is expensive

A student streaming lessons burns more money than the phone costs. Preloaded data is limited and metered.



Power is unreliable

Lessons must survive a blackout mid-question and sync later, without losing a single attempt.



One-size-fits-all prep

Past questions are dumped as PDFs. Nobody tells a learner which topic is actually costing them marks.



Paywalls exclude

Paid-only study apps lock out the students who need them most. A scholarship pool should cover them.

THE SOLUTION

EduBridge AI — one PWA, three guarantees

Offline-first by architecture, adaptive by measurement, and free at the core.

1 Works offline, always

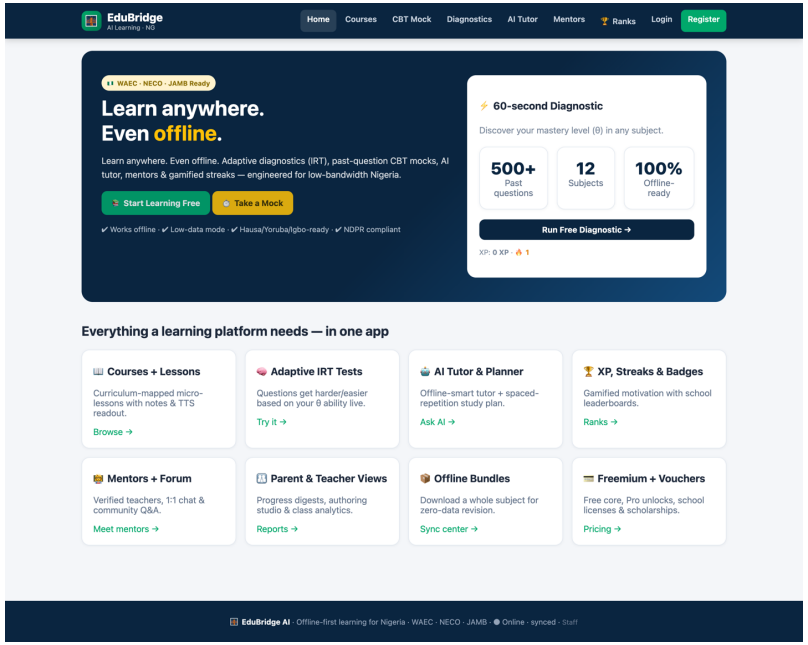
Service worker precache + IndexedDB + delta-sync queue. Every attempt, XP event and message queues locally and reconciles on reconnect.

2 Adapts to the learner

A real IRT engine estimates ability (theta) from every answer and drives mastery bars, weak-topic detection and a spaced-repetition study plan.

3 Free core, fair monetisation

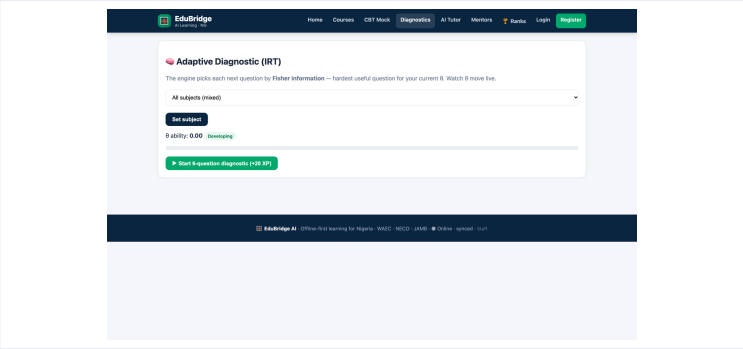
Courses, CBT mocks and a daily AI tutor stay free. Paystack/Flutterwave, vouchers and a scholarship pool monetise schools without excluding pupils.



PRODUCT TOUR

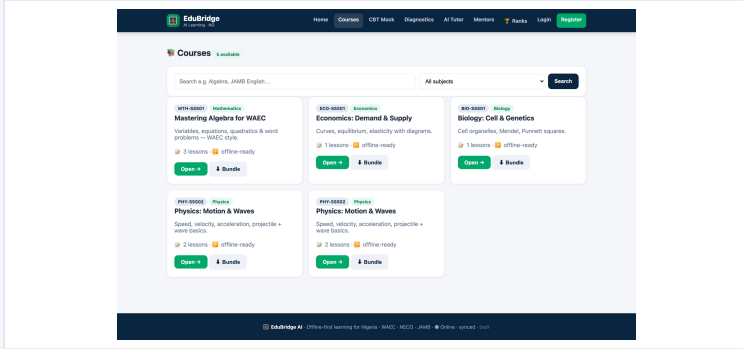
One app, the full exam-prep loop

Diagnose → learn → practise → adapt → prove.



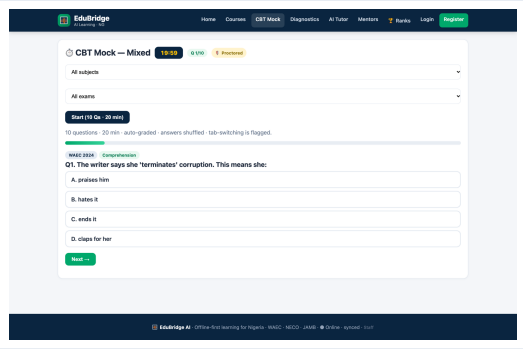
Adaptive diagnostics

IRT ability estimate + weak-topic detection



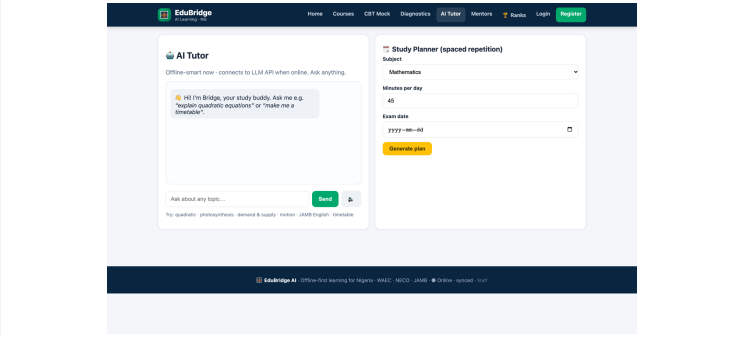
Course catalogue

WAEC / NECO / JAMB mapped, offline bundles



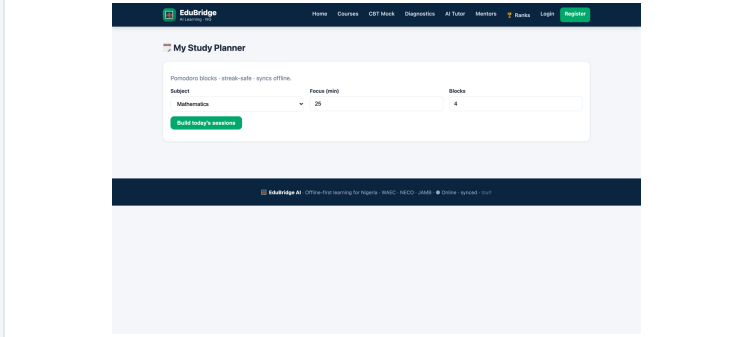
Timed CBT mock

Auto-grading, shuffle, timer, tab-blur flag



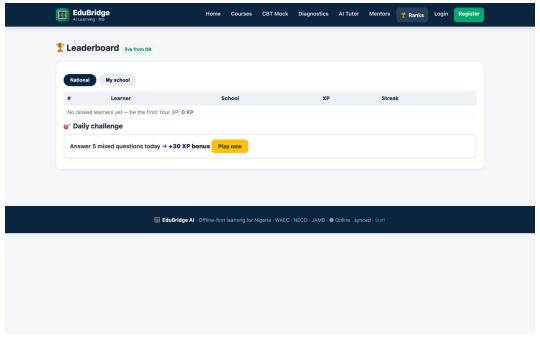
AI tutor

Offline rule-based answers, 60 msgs/day on Medium



Study planner

Spaced repetition around each learner's gaps



Gamification

XP, streaks, badges, ranks, daily challenge

ARCHITECTURE

Deploy on a school server today. Scale tomorrow.

Zero build step: PHP 8 + MySQL on Apache. No npm, no bundler, no lock-in.

CLIENT · PWA

Vanilla SPA shell · IndexedDB (courses, quiz, queue) · Service worker precache · Offline bundles · Low-data mode · TTS · i18n (EN/HA/YO/IG/Pidgin)

API GATEWAY

TLS · rate limiting · router · session + JWT · RBAC (student / teacher / parent / school_admin / admin)

BACKEND SERVICES · PHP APIs

A Diagnostic & Sync · B IRT Assessment · C Mentorship & Chat · D AI Tutor & Planner · E Content & Live · F Engagement · G Payments & Enrollment · H Analytics & Safety

DATA

MySQL (users, courses, attempts, xp) · JSON seed bundles · IndexedDB mirror · Redis (later) · R2/S3 media · notification queue (mail / SMS / USSD)

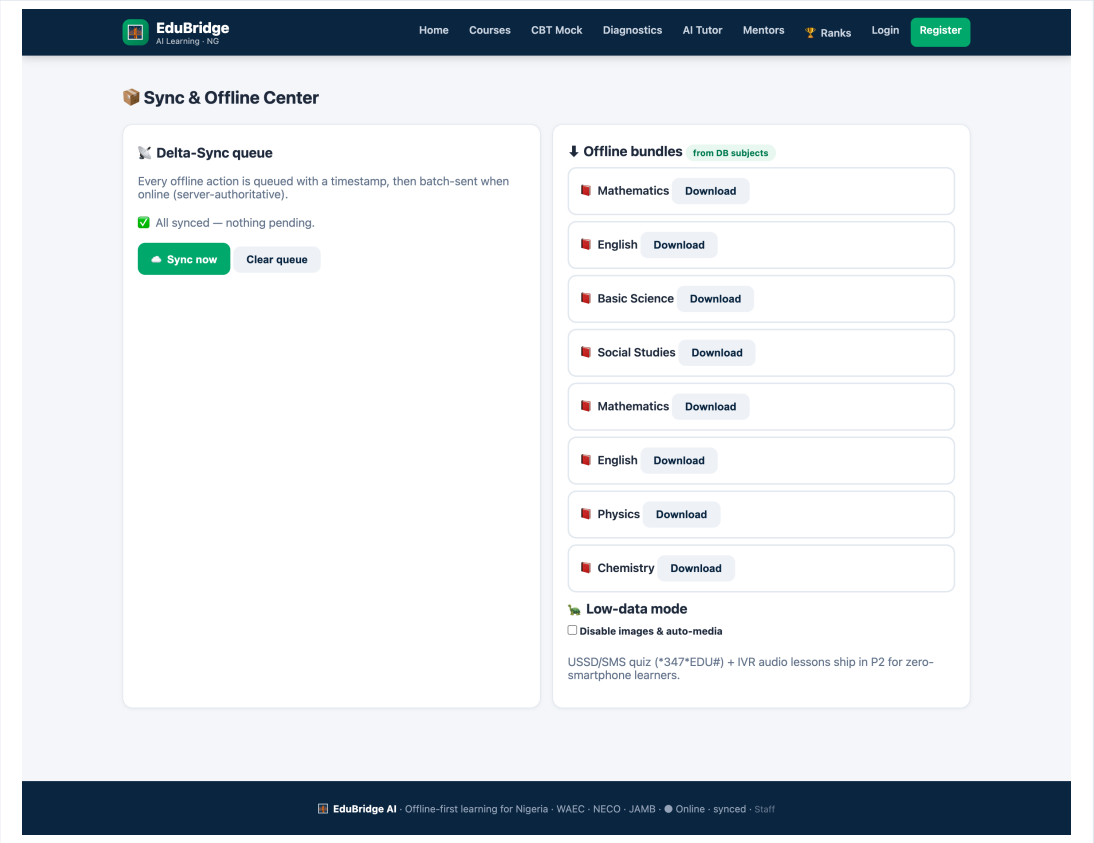
12 SUBSYSTEMS

- LMS Core
- Question Bank & CBT
- Adaptive IRT
- AI Tutor / Planner
- Gamification
- Live Class & Mentorship
- Community
- Authoring Studio
- Analytics (4 roles)
- Offline & Low-data
- Payments & Scholarships
- Safety & Compliance

DIFFERENTIATOR

Offline-first is not a feature — it is the architecture

Most apps are online with an offline screen. EduBridge is offline with an online sync.

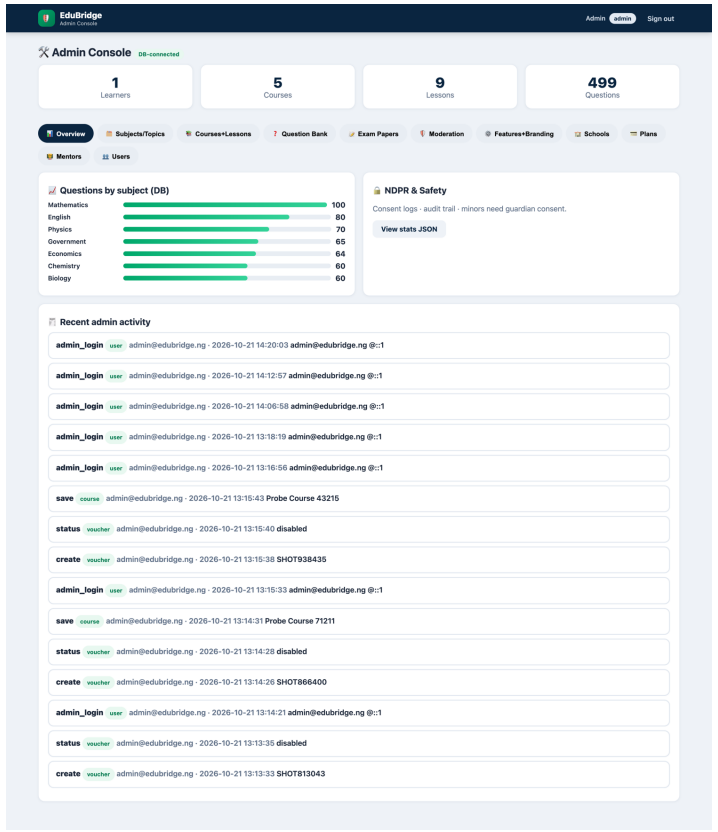


- **Precache the shell** Service worker keeps the app instant and available with zero network.
- **Local-first writes** Attempts, XP, messages and mentor chats land in IndexedDB immediately.
- **Delta-sync queue** Every mutation is queued with an op + payload and replayed on reconnect — idempotently.
- **Low-data mode** Learners choose a byte budget; bundles download once over WiFi and never again.
- **Offline bundle downloader** Courses, questions and media cached per subject for true 2G use.

OPERATOR SIDE

A standalone staff console ships with the product

Content, moderation, commerce and people — with its own session namespace and audit trail.



Content

Subjects, topics, courses, lessons, question bank, exam papers, moderation queue

Commerce

Plans & pricing, vouchers, bulk enrolment, school licenses, payments

People

Users & roles, schools, mentors, guardians and parent links

Platform

Feature flags, branding, settings — and live audit logging of every action

Separate session namespace · bcrypt auth · brute-force lockout · every action audited

BUSINESS MODEL

Free core, monetised at the edges

The free tier is the growth engine; schools and power users fund the platform.

Free

₦0

forever

- Courses
- CBT mocks
- AI tutor (daily limit)
- Offline bundles

Medium

₦500

per term

- Everything in Free
- 60 AI messages/day
- Study planner
- Forum

MOST POPULAR

Pro

₦1,500

per term

- Unlimited AI
- Adaptive plans
- Verified mentors
- Certificates
- Parent SMS · No ads

School

Custom

contact sales

- Teacher studio
- Class analytics
- Bulk vouchers
- Scholarships

Payments: Paystack / Flutterwave + bank transfer + vouchers. No learner is locked out — the scholarship pool covers indigent students.

ROADMAP

Eight weeks to a two-school pilot

P0 is shipped. The next two pushes close the loop from 'good app' to 'institutional product'.

Wk 1–2

P0 polish + seed 500 real past questions

Harden the 499-question bank with authentic WAEC/NECO/JAMB items and tighten the IRT calibration.

Wk 3–4

Live classes + payments sandbox + i18n

WebRTC/Jitsi live classes with attendance, Paystack sandbox checkout, and a Hausa/Yoruba/Igbo/Pidgin toggle.

Wk 5–6

Authoring studio + deep analytics

Teacher course authoring with lesson drafts, plus per-role analytics dashboards and an at-risk churn predictor.

Wk 7–8

Pilot in 2 schools + hardening + NDPR audit

Supervised rollout, consent and data-protection audit, offline soak tests on real 2G, and certificate delivery.

THE ASK

Put every Nigerian learner's exam prep in their pocket — for free, offline.

Pilot — Two schools, 8-week supervised rollout with offline soak testing on real 2G.

Content — Access to authentic WAEC/NECO/JAMB past papers to seed the bank past 500 items.

Partners — Payment rails (Paystack/Flutterwave) and SMS gateways for parent digests.